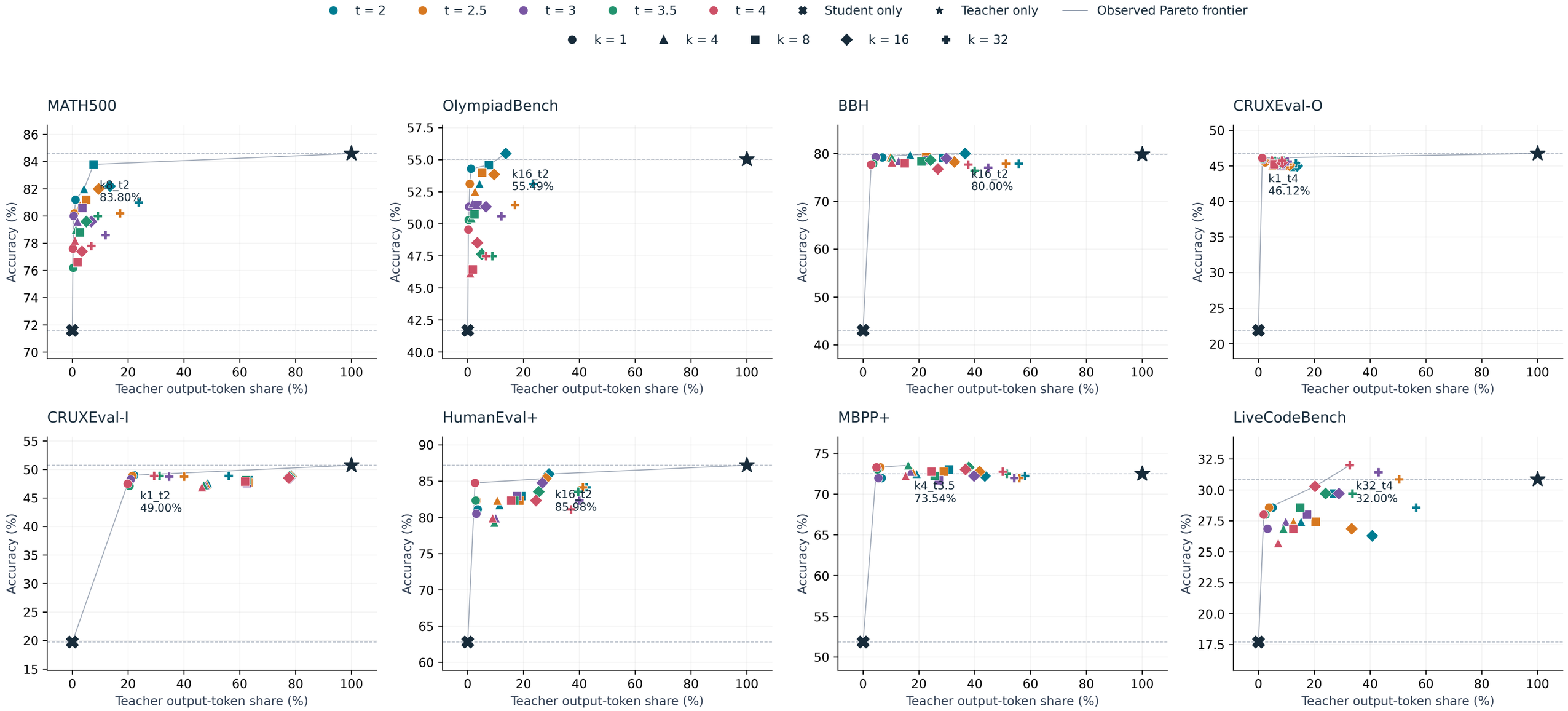
Accuracy vs. teacher token share



Qwen3-1.7B → Qwen3-14B | Each point is one configuration. Axes differ by benchmark.
Token share excludes prompts / discarded drafts; it is not latency or total compute cost. Stopping rules differ; no uncertainty estimates supplied.

Accuracy (%)

	MATH500	OlympiadBench	BBH	CRUXEval-O	CRUXEval-I	HumanEval+	MBPP+	LiveCodeBench
student_only	71.60	41.69	43.06	21.88	19.75	62.80	51.85	17.71
teacher_only	84.60	55.04	79.81	46.75	50.75	87.20	72.49	30.86
k1_t2	81.20	54.30	79.17	45.50	**49.00**	81.10	71.96	28.57
k4_t2	82.00	53.12	79.72	45.12	47.62	81.71	72.49	27.43
k8_t2	**83.80**	54.60	79.07	44.88	48.12	82.93	73.02	29.71
k16_t2	82.20	**55.49**	**80.00**	45.00	48.88	**85.98**	72.22	26.29
k32_t2	81.00	53.12	77.87	45.38	48.88	84.15	72.22	28.57
k1_t2.5	80.20	53.12	79.17	45.50	48.88	82.32	73.28	28.57
k4_t2.5	80.80	52.52	79.26	45.12	47.38	82.32	72.75	27.43
k8_t2.5	81.20	54.01	79.26	45.00	47.75	82.32	72.75	27.43
k16_t2.5	82.00	53.86	78.24	45.12	48.75	85.37	72.75	26.86
k32_t2.5	80.20	51.48	77.87	45.25	48.75	84.15	71.96	30.86
k1_t3	80.00	51.34	79.26	45.88	48.25	80.49	71.96	26.86
k4_t3	79.60	51.63	78.43	45.75	47.25	79.88	72.75	27.43
k8_t3	80.60	51.48	78.61	45.12	47.62	82.93	71.69	28.00
k16_t3	79.60	51.34	78.98	45.50	48.50	84.76	72.22	29.71
k32_t3	78.60	50.59	77.04	45.62	48.75	82.32	71.96	31.43
k1_t3.5	76.20	50.30	77.87	46.00	47.12	82.32	73.02	28.00
k4_t3.5	79.00	50.45	79.17	45.88	47.25	79.27	**73.54**	26.86
k8_t3.5	78.80	50.74	78.33	45.38	48.12	82.32	72.22	28.57
k16_t3.5	79.60	47.63	78.61	45.50	48.75	83.54	73.28	29.71
k32_t3.5	80.00	47.48	76.39	45.75	48.88	83.54	72.49	29.71
k1_t4	77.60	49.55	77.69	**46.12**	47.50	84.76	73.28	28.00
k4_t4	78.20	46.14	78.15	46.00	46.88	79.88	72.22	25.71
k8_t4	76.60	46.44	77.96	45.25	47.88	82.32	72.75	26.86
k16_t4	77.40	48.52	76.76	45.50	48.50	82.32	73.02	30.29
k32_t4	77.80	47.48	77.69	45.75	48.88	81.10	72.75	**32.00**

Cell values are raw measurements; color is scaled independently within each benchmark column.
Darker = larger (not always better). Bold accuracy = best hybrid configuration within that benchmark.
Stopping rules / baseline generation differ. See source.md for sample counts and full methodology.

Mean interventions

	MATH500	OlympiadBench	BBH	CRUXEval-O	CRUXEval-I	HumanEval+	MBPP+	LiveCodeBench
student_only	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
teacher_only	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
k1_t2	9.71	17.94	18.61	2.56	4.07	6.53	6.13	28.93
k4_t2	8.67	15.54	13.09	1.44	2.21	5.11	4.55	21.90
k8_t2	8.00	14.17	10.25	1.19	1.61	4.34	3.79	17.83
k16_t2	7.46	12.90	7.49	1.04	1.21	3.39	2.76	13.34
k32_t2	6.09	10.48	5.24	1.00	1.02	2.58	2.01	9.69
k1_t2.5	6.29	10.96	15.78	2.52	3.95	5.74	5.80	22.47
k4_t2.5	5.72	9.78	11.37	1.42	2.19	4.85	4.45	17.67
k8_t2.5	5.16	9.39	9.01	1.22	1.60	4.27	3.55	14.25
k16_t2.5	4.79	8.69	6.79	1.10	1.21	3.27	2.70	11.78
k32_t2.5	4.50	7.50	4.86	1.00	1.01	2.52	1.95	8.91
k1_t3	4.41	7.33	13.80	2.48	3.82	5.55	5.19	18.18
k4_t3	3.83	6.57	10.05	1.41	2.17	4.56	4.15	14.95
k8_t3	3.69	6.47	8.12	1.21	1.59	4.02	3.43	12.67
k16_t3	3.46	6.00	6.22	1.10	1.21	3.16	2.61	9.36
k32_t3	3.22	5.36	4.56	1.00	1.01	2.48	1.92	7.81
k1_t3.5	3.08	5.38	12.12	2.52	3.73	5.12	4.92	14.44
k4_t3.5	2.69	4.76	9.07	1.40	2.15	4.32	4.02	11.93
k8_t3.5	2.76	4.62	9.34	1.20	1.59	3.77	3.24	10.41
k16_t3.5	2.65	4.47	5.66	1.10	1.21	3.02	2.52	8.57
k32_t3.5	2.52	4.04	4.25	1.00	1.01	2.45	1.88	6.55
k1_t4	2.15	3.76	10.94	2.39	3.62	4.62	4.69	11.63
k4_t4	2.12	3.51	8.15	1.40	2.13	3.99	3.91	10.30
k8_t4	2.06	3.55	6.74	1.15	1.58	3.50	3.11	8.83
k16_t4	1.83	3.28	5.27	1.10	1.21	2.88	2.49	7.35
k32_t4	1.85	2.94	4.03	1.00	1.01	2.27	1.81	6.07

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Stopping rules / baseline generation differ. See source.md for sample counts and full methodology.

Teacher output-token share (%)

	MATH500	OlympiadBench	BBH	CRUXEval-O	CRUXEval-I	HumanEval+	MBPP+	LiveCodeBench
student_only	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
teacher_only	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
k1_t2	1.15	1.19	6.91	2.41	22.14	3.58	6.68	5.09
k4_t2	4.20	4.23	16.87	6.01	48.55	11.41	19.16	15.25
k8_t2	7.63	7.58	28.71	12.48	63.22	19.18	30.76	27.02
k16_t2	13.52	13.66	36.55	13.85	78.42	29.10	43.63	40.74
k32_t2	23.83	23.36	55.79	13.40	56.04	42.57	58.01	56.49
k1_t2.5	0.76	0.74	4.65	2.36	21.54	3.18	6.17	3.78
k4_t2.5	2.83	2.62	9.86	4.96	48.08	10.63	17.81	12.54
k8_t2.5	4.95	5.16	22.61	9.33	63.04	18.50	28.88	20.46
k16_t2.5	9.42	9.48	32.70	11.08	78.34	28.00	41.72	33.40
k32_t2.5	17.10	16.95	51.17	10.46	40.05	41.26	55.95	50.39
k1_t3	0.52	0.49	4.54	1.57	20.91	3.09	5.49	3.22
k4_t3	1.86	1.83	12.74	7.04	47.61	10.08	16.94	9.78
k8_t3	3.62	3.50	23.14	8.06	62.54	17.71	27.14	17.38
k16_t3	6.81	6.54	29.85	9.09	77.96	26.71	39.76	28.79
k32_t3	11.93	12.09	44.81	10.46	34.69	39.99	54.14	43.00
k1_t3.5	0.35	0.37	3.60	1.45	20.48	2.86	5.12	2.49
k4_t3.5	1.29	1.30	10.34	5.96	47.13	9.58	16.18	8.94
k8_t3.5	2.69	2.41	20.92	6.20	62.31	16.73	25.66	14.93
k16_t3.5	5.02	5.03	24.15	7.96	77.81	25.45	37.86	24.04
k32_t3.5	9.15	8.86	39.96	8.88	31.26	39.58	51.48	33.64
k1_t4	0.26	0.25	2.87	1.35	19.83	2.62	4.80	1.87
k4_t4	0.95	0.91	10.34	4.83	46.46	8.96	15.25	7.06
k8_t4	1.89	1.84	14.93	5.66	61.92	15.60	24.42	12.47
k16_t4	3.46	3.45	26.79	8.48	77.58	24.46	36.70	20.25
k32_t4	6.83	6.59	37.67	8.43	29.33	37.02	50.05	32.70

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Mean output tokens

	MATH500	OlympiadBench	BBH	CRUXEval-O	CRUXEval-I	HumanEval+	MBPP+	LiveCodeBench
student_only	885.08	1604.45	479.75	402.01	349.06	199.18	139.05	618.22
teacher_only	911.39	1524.58	279.43	115.12	18.23	177.63	90.15	546.82
k1_t2	842.03	1510.33	269.08	106.33	18.36	182.60	91.89	567.96
k4_t2	824.53	1458.84	303.92	95.77	18.16	178.95	92.10	572.71
k8_t2	837.39	1482.34	274.39	68.52	18.20	180.31	91.52	524.41
k16_t2	879.40	1494.41	306.16	79.81	18.14	181.36	89.21	516.69
k32_t2	808.50	1409.54	268.25	102.32	27.58	178.18	91.13	536.02
k1_t2.5	829.13	1479.97	339.22	106.59	18.35	180.76	94.10	593.76
k4_t2.5	807.28	1482.64	451.22	114.70	18.24	182.20	97.02	561.63
k8_t2.5	830.83	1440.49	304.97	94.43	18.17	183.64	91.12	554.13
k16_t2.5	809.57	1441.71	306.82	108.90	18.14	181.47	90.38	555.95
k32_t2.5	832.38	1383.82	268.37	131.78	38.53	179.35	91.35	552.02
k1_t3	841.23	1482.96	304.23	158.34	18.28	179.61	94.48	563.92
k4_t3	823.16	1414.21	307.83	80.15	18.22	180.81	94.77	609.34
k8_t3	814.26	1456.42	267.48	108.80	18.21	180.92	93.68	580.23
k16_t3	808.41	1438.13	306.75	132.74	18.21	183.88	92.11	513.47
k32_t3	854.73	1377.94	284.82	132.49	44.45	181.65	93.89	570.83
k1_t3.5	866.49	1462.08	336.78	174.12	18.23	179.32	96.02	580.63
k4_t3.5	834.26	1443.61	341.02	93.44	18.21	180.15	96.07	531.71
k8_t3.5	818.37	1505.72	342.84	140.31	18.20	179.65	93.10	554.88
k16_t3.5	838.94	1387.82	341.50	151.42	18.18	184.75	93.36	562.46
k32_t3.5	868.77	1406.15	292.69	155.54	49.26	181.82	96.48	609.21
k1_t4	822.84	1523.21	380.88	177.48	18.25	176.31	97.59	620.26
k4_t4	885.79	1504.16	305.89	115.36	18.30	177.83	99.31	582.14
k8_t4	869.41	1515.12	341.00	146.33	18.27	178.57	94.14	563.57
k16_t4	840.68	1485.23	284.98	142.72	18.21	183.10	94.97	574.39
k32_t4	854.41	1379.40	292.08	162.62	52.49	181.39	95.24	580.33

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